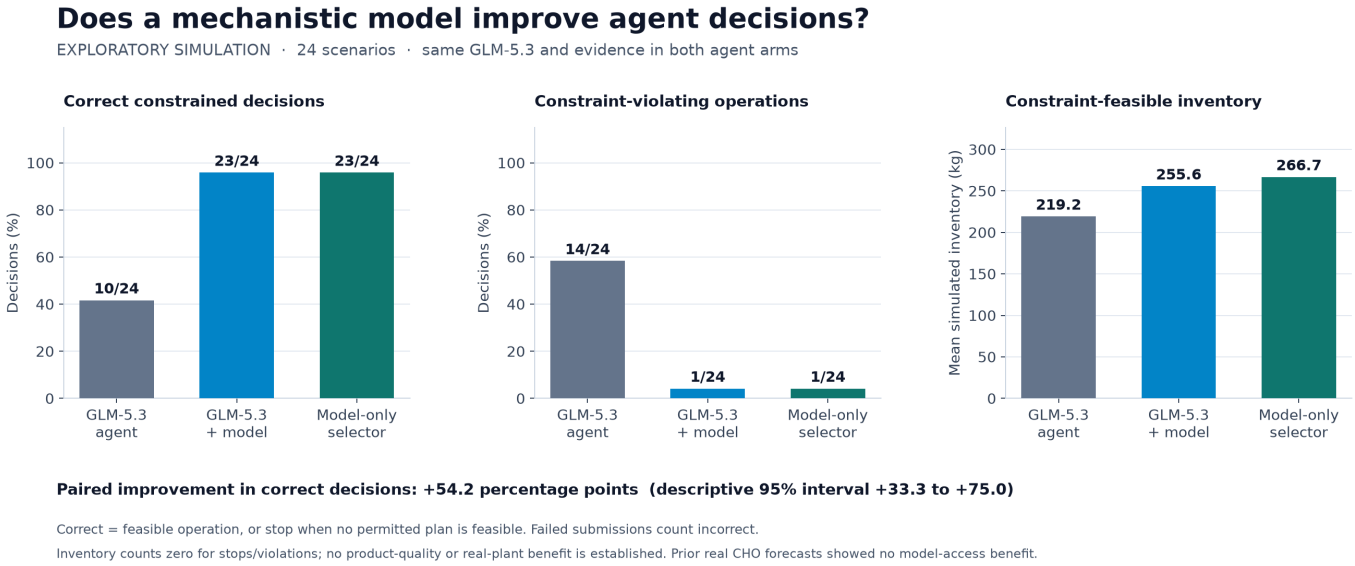
Sievance | Agents + mechanistic models

Exploratory evaluation | November 1, 2026 | Actual Modal GLM-5.3 endpoint

Model access improved constrained decisions in this simulated cohort.

24 independent simulated scenarios; 48 planned agent sessions. 48 completed submissions. The same evidence and decision policy were used in both arms.



Measure	Agent alone	Agent + model
Correct constrained decisions	10/24	23/24
Constraint-violating operations	14	1
Correct / unnecessary stops	0 / 0	12 / 0
Mean simulated inventory (kg)	323.45	266.48
Mean constraint-feasible inventory (kg)	219.21	255.63

Paired correct-decision difference: **+54.2 percentage points**; descriptive 95% interval [+33.3, +75.0]. Scenarios, not model calls, are the independent units.

Interpretation: The conventional model-only selector also scored 23/24; added agent value over that selector was not demonstrated. These are synthetic process outcomes using a shared mechanistic equation family, including deliberately mismatched receiving conditions. They do not establish real manufacturing performance, product quality, or avoided losses. The earlier real CHO forecast comparison did not show benefit from the selected model.

Correct means feasible operation when a permitted plan is feasible, or stopping otherwise. Failed submissions count incorrect. Constraint-feasible inventory is zero for violations/stops; it is not measured product quality or released yield.

A prior cost pilot was stopped before comparative analysis and is excluded. New scenario seeds and analysis were frozen before inference. A conventional model-only comparator and every case are in the full report.

Protocol SHA-256: 1f0e2618a9b9634eeed23478... | 98 forwarded model requests. Costs are token-rate estimates/reservations, not a verified invoice.